3.2.4 Population and the environment

This optional section of our specification has been designed to explore the relationships between key aspects of physical geography and population numbers, population health and well-being, levels of economic development and the role and impact of the natural environment. Engaging with these themes at different scales fosters opportunities for students to contemplate the reciprocating relationships between the physical environment and human populations and the relationships between people in their local, national and international communities.

Study of this section offers the opportunity to exercise and develop observation skills, measurement and geospatial mapping skills, together with data manipulation and statistical skills, including those associated with and arising from fieldwork.

3.2.4.1 Introduction

The environmental context for human population characteristics and change. Key elements in the physical environment: climate, soils, resource distributions including water supply. Key population parameters: distribution, density, numbers, change. Key role of development processes. Global patterns of population numbers, densities and change rates.

3.2.4.2 Environment and population

Global and regional patterns of food production and consumption. Agricultural systems and agricultural productivity. Relationship with key physical environmental variables – climate and soils.

Characteristics and distribution of two major climatic types to exemplify relationships between climate and human activities and numbers. Climate change as it affects agriculture.

Characteristics and distribution of two key zonal soils to exemplify relationship between soils and human activities especially agriculture. Soil problems and their management as they relate to agriculture: soil erosion, waterlogging, salinisation, structural deterioration.

Strategies to ensure food security.

3.2.4.3 Environment, health and well-being

Global patterns of health, mortality and morbidity. Economic and social development and the epidemiological transition.

The relationship between environment variables eg climate, topography (drainage) and incidence of disease. Air quality and health. Water quality and health.

The global prevalence, distribution, seasonal incidence of one specified biologically transmitted disease, eg malaria; its links to physical and socio-economic environments including impacts of environmental variables on transmission vectors. Impact on health and well-being. Management and mitigation strategies.

The global prevalence and distribution of one specified non-communicable disease, eg a specific type of cancer, coronary heart disease, asthma; its links to physical and socio-economic environment including impacts of lifestyles. Impact on health and well-being. Management and mitigation strategies.

Role of international agencies and NGOs in promoting health and combating disease at the global scale.

3.2.4.4 Population change

Factors in natural population change: the demographic transition model, key vital rates, age–sex composition; cultural controls. Models of natural population change, and their application in contrasting physical and human settings. Concept of the Demographic Dividend.

International migration: refugees, asylum seekers and economic migrants: environmental and socio-economic causes, processes. Demographic, environmental, social, economic, health and political implications of migration.

3.2.4.5 Principles of population ecology and their application to human populations

Population growth dynamics. Concepts of overpopulation, underpopulation and optimum population. Implications of population size and structure for the balance between population and resources; the concepts of ‘carrying capacity’ and ‘ecological footprint’ and their implications.

Population, resources and pollution model: positive and negative feedback. Contrasting perspectives on population growth and its implications; Malthusian, neo-Malthusian and alternatives such as associated with Boserup and Simon.

3.2.4.6 Global population futures

Health impacts of global environmental change: ozone depletion – skin cancer, cataracts; climate change – thermal stress, emergent and changing distribution of vector borne diseases, agricultural productivity and nutritional standards.

Prospects for the global population. Projected distributions. Critical appraisal of future population-environment relationships.

3.2.4.7 Case studies

Case study of a country/society experiencing specific patterns of overall population change – increase or decline – to illustrate and analyse the character, scale, and patterns of change, relevant environmental and socio-economic factors and implications for the country/society.

Case study of a specified local area to illustrate and analyse the relationship between place and health related to its physical environment, socio-economic character and the experience and attitudes of its populations.